Current Process for Including the New Rotation Term

Derivation of the Rotational Paraxial Wave Equation from the Exact Envelope Equation

Let's say we have a wave propagating in z, in a frame rotating about y such that $\mathbf{\Omega} = (0, \Omega, 0)$. As taken from the GPT-aided derivation, the expanded version of the exact envelope equation is:

$$
\begin{aligned}
c^2\left(A_{XX} + A_{YY} + A_{ZZ} + 2ikA_Z - k^2 A\right) = A_{\tau\tau} &+ (-2i\omega + 2ik\Omega X)A_\tau \\
&+ (-\omega^2 + 2\omega k\Omega X - k^2\Omega^2 X^2 - ik\Omega^2 Z)A \\
&+ 2\Omega(XA_{\tau Z} - ZA_{\tau X}) \\
&+ 2i\omega\Omega(ZA_X - XA_Z) \\
&+ \Omega^2(Z^2 A_{XX} + X^2 A_{ZZ} - 2XZA_{XZ}) \\
&- \Omega^2(XA_X + ZA_Z) \\
&- ik\Omega^2 ZA - ik\Omega^2 XZA_X + ik\Omega^2 X^2 A_Z \\
&- 2ik\Omega^2 XZA_X + 2ik\Omega^2 X^2 A_Z.
\end{aligned}
$$

In order to simplify the equation, we make the slowly varying envelope approximation (SVEA), and drop the following terms:

- Any term that includes $A_{\tau\tau}$, A_{ZZ}, or $A_{\tau Z}$ because the E-field varies slowly in time and z-direction compared to how it evolves with other variables (for example, in the transverse directions x and y).
- Any term that includes Ω^2 because Ω is some very small number (e.g. $\Omega \approx 0.001$ rad/s).
- The $-c^2k^2A$ term on the LHS because: $\omega^2 = c^2k^2$, which cancels with the $-\omega^2 A$ term from the RHS.

We are now left with:

$$
\begin{aligned}
c^2\left(A_{XX}+A_{YY}+2ikA_Z\right) = & -2i\omega A_\tau + 2ik\Omega X A_\tau \\
& + 2\omega k\Omega X A - 2\Omega Z A_{\tau X} \\
& + 2i\omega\Omega Z A_X - 2i\omega\Omega X A_Z \\
& + 2ik\Omega X A_\tau
\end{aligned}
$$

Inserting $\omega = ck$ and rearranging/combining:

$$
-2ickA_\tau + 2ik\Omega X A_\tau - 2c^2 ikA_Z - 2ick\Omega X A_Z = c^2\nabla_\perp^2 A - 2ck^2\Omega X A - 2\Omega A_{\tau X} - 2ick\Omega Z A_X
$$

Dividing by $-2ic^2k$,

$$
\left[\frac{1}{c}\frac{\partial A}{\partial \tau} + \frac{\partial A}{\partial Z}\right] = \frac{k\Omega X A}{ic} + \frac{\Omega}{ikc^2}A_{\tau X} + \frac{\Omega}{c}ZA_X + \frac{1}{2ik}\nabla_\perp^2 A
$$

Good things:

- We recover the expected LHS, as well as the rotation "shearing" term, and the usual diffraction term with the Laplacian squared.

Things I'm still questioning:

- We can probably drop the mixed second derivative $\frac{\Omega}{ikc^2}A_{\tau X}$? It is much smaller than the other terms and I know that we didn't include it in our previous discussions. But for now, I kept it here for completion.
- I originally had the terms $2ik\Omega X A_\tau$ and $2ik\Omega X A_Z$ on the LHS (see above) with the rest of the differential operator (complete derivative). I dropped them because I think that they either cancel out ($A_\tau = A_Z$?) or they are somehow negligible.
- Signage. I may have missed some minus signs. I think that the RHS maybe is supposed to all be negative.

Note that in my formulation, the new term actually has the form:

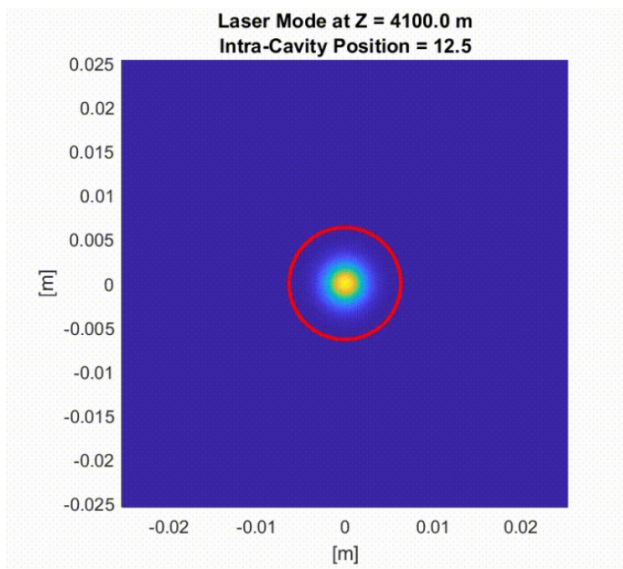
$$
\frac{k\Omega X A}{ic}
$$

NOT

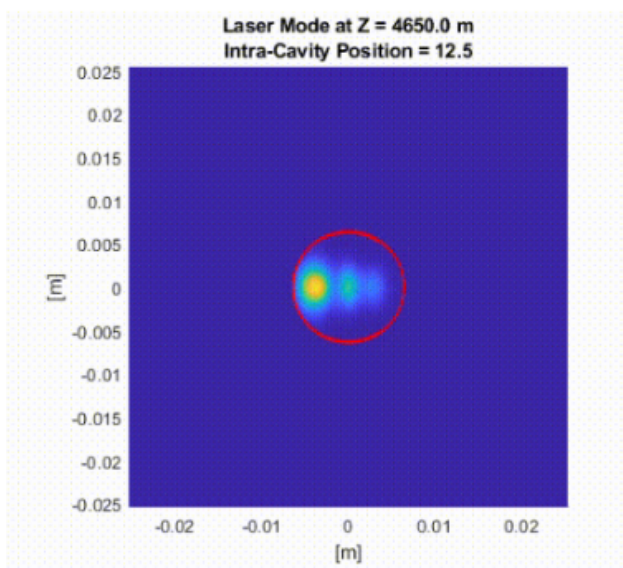
$$\frac{-2k^2\Omega XA}{c}$$

It lost its prefactor $-2ck^2$ when we divided through by $-2ic^2k$.

I've tested this new form in the code and its effect on the laser mode is much less extreme than the k^2 version. With an $\Omega = 0.001$ rad/s, this new form does not significantly deform the mode and the mode is able to converge to a steady state with a single intensity lobe:



For comparison, the $\frac{-2k^2\Omega XA}{c}$ version "steady-state" looked like this with $\Omega = 0.001$ rad/s:



Practically, I implement this term as a phase screen in real space with the form:

$$\exp\left(\frac{k\Omega X}{ic}dz\right)$$

Which comes from solving the equation by separation of variables.

0.1 2nd Pass Analysis

$$\left[\frac{1}{c}\frac{\partial A}{\partial \tau}+\frac{\partial A}{\partial Z}\right]=\frac{k\Omega XA}{ic}+\frac{\Omega}{c}ZA_X+\frac{1}{2ik}\nabla_\perp^2 A$$

$$\left[\frac{1}{c}\frac{\partial E}{\partial \tau}+\frac{\partial E}{\partial Z}\right]=\frac{i}{2k}\nabla_\perp^2 E$$